

Examining Gender Differences in the Roles of Meaning in Life and Interpersonal Expectancies in Depressive Symptoms

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Abstract In the present study, we examined for gender differences in meaning in life (viz., presence of and searching for) and interpersonal expectancies (viz., positive and negative) as additive and interactive predictors of depressive symptoms in a sample of 117 male and 132 female college students. We found that meaning in life accounted for a significant amount of variance in depressive symptoms for both men and women. While presence of meaning in life was a significant predictor of depressive symptoms for both groups, it was a stronger predictor among women compared to men. Furthermore, we found that interpersonal expectancies accounted for additional variance in depressive symptoms above and beyond meaning in life for both groups, and interestingly, only positive interpersonal expectancy emerged as a significant predictor of depressive symptoms. Finally, the interactions involving meaning in life and interpersonal expectancies accounted for a significant amount of additional variance above and beyond the main effects for both men and women. However, a greater number of significant interactions emerged for women than for men. Implications of the present findings for understanding gender differences in meaning in life and interpersonal expectancies as predictors of depressive symptoms are discussed.

Keywords Meaning in life · Interpersonal expectancies · Depressive symptoms · Gender differences

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Introduction

Meaning in life can be understood to be an individuals' sense of his/her existence (e.g., [48, 52, 56]). One of the quintessential characteristics that define human beings is our innate desire to understand ourselves and the world in which we live. According to existential scholars and thinkers, this passion is most clearly manifested in our strivings to attain meaning in our lives (e.g., [5, 9, 46]). More than a century ago, James [28] highlighted the importance of understanding the meaning of life in his essay, "*What Makes a Life Significant.*" Similarly, in *Man's Search for Meaning*, Frankl [21] underscored the positive power associated with searching for meaning in one's life as a way to transcend tragic moments of human co-existence. In the past decade, psychology researchers have regained an interest in the empirical study of meaning in life. Areas of study have ranged from how meaning in life is constructed from life stressors, to how meaning in life changes throughout the lifetime (e.g., [23, 50]). Additionally, psychologists have also gained interest in studying the importance of one's meaning in life in relation to adjustment (see [41] for review). In that regard, findings from studies using various measures of meaning in life have generally shown that it is reliably associated with a wide range of outcomes, including life satisfaction (e.g., [17, 40]), happiness (e.g., [24, 26]), positive affect (e.g., [24, 25]), psychological well-being (e.g., [44, 45]), stress (e.g., [18, 36]), depression (e.g., [51]), and suicide risk (e.g., [20, 33]). Overall, the pattern that has emerged from past studies indicates that greater meaning in life is positively associated with positive outcomes (e.g., life satisfaction, psychological well-being), and negatively associated with negative outcomes (e.g., depression, suicide risk).

However, most research on meaning in life has focused on the general experience/perception of meaning in life, with little regard to the construct of searching for meaning in life. The Meaning in Life Questionnaire (MLQ; [49]), which is one of the most highly appraised instruments to measure meaning in life (see [8]), differentiates the presence of meaning in life from searching for meaning in life. *Presence of meaning in life* is defined as "the extent to which people comprehend, make sense of, or see significance in their lives, accompanied by the degree to which they perceive themselves to have a purpose, mission, or overarching aim in life," [48] and *searching for meaning in life* is defined as "the strength, intensity, and activity of people's desire and efforts to establish and/or augment their understanding of meaning, significance, and purpose of their lives" [50]. In studies that have examined for dimensions of meaning in life, presence of meaning in life has generally been indicated to have a more consistent and adaptive function compared to searching for meaning in life. For example, in a large multi-age sample of adults (ages ranging from 18 to those over 65), [51] found that presence of meaning in life was consistently and negatively associated with depressive symptoms, whereas searching for meaning in life was positively associated with depressive symptoms. Thus, presence of meaning in life appears to involve more positive adjustment conditions, whereas searching for meaning in life appears to involve challenges that might hinder one's ability to attain desired positive adjustment conditions.

Yet, perhaps just as intriguing as the findings regarding presence of, and searching for, meaning in life, Steger and his colleagues [51] also reported findings of significant gender differences in both dimensions of meaning in life across their multi-age sample of adults. Specifically, these investigators found that women reported higher levels of presence of, and searching for, meaning in life than men. This may be explained by the finding that in early to middle adulthood, women, compared to men, express greater aspirations for, and attainment of, major life goals (e.g., good marriage, having children, obtaining an interesting job, etc.; [42]), and thus may lead to a greater efforts to search for, and later perceive, meaning in life. However, beyond examining mean differences in meaning in life scores between men and women, researchers have yet to extend studies to examine for potential gender differences in the function of meaning in life. Specifically, one question that remains unaddressed is how presence of, and searching for, meaning in life may be differentially associated with depressive symptoms for men and women. For example, given that women tend to have higher scores on both dimensions of meaning in life compared to men, it is not clear whether the negative association between presence of meaning in life and depressive symptoms, or the positive association between searching for meaning in life and depressive symptoms, found by Steger et al. [51] is stronger for women compared to men, or similar for both groups. Clarifying this relationship would be beneficial for identifying the group/s that may benefit the most from the use of meaning-based interventions (e.g., existential psychotherapy, logotherapy; [22, 56]). Given that women may encounter unique challenges that make perceiving and searching for meaning more difficult compared to men (e.g., oppression, sexism, violence, miscarriage; [16, 39]), it may be that meaning in life is a stronger predictor of depressive symptoms for women than for men. Therefore, gaining a better understanding of how this relationship may differ between men and women has implications for future research in the study of meaning in life and adjustment as well as clinical implications for the development of more specified interventions involving meaning in life for men and women.

In addition to the possible gender differences that may exist in the associations between meaning in life and depressive symptoms, it would be useful to determine whether other factors, specifically interpersonal factors, may buffer and/or exacerbate these associations. Interpersonal factors are important determinants of depressive symptoms (e.g., [10, 12, 27]). According to a review by Kawachi and Berkman [31], women's mental health may be more greatly influenced by interpersonal factors compared to men. Some have argued that women, compared to men, are more strongly motivated by self-schemas centered on relationships (i.e., the relational self; [29, 54]). Indeed, findings from studies over the past two decades have indicated that women are strongly motivated to maintain close relationships with others (e.g., [7, 13, 30, 35]). For example, in a study of nineteen different life goals, Abowitz and Knox [1] found that undergraduate women rated the goals of having close friends and having a lot of family/relatives as significantly more important compared to the ratings by undergraduate men. Additionally, interpersonal relationships are often identified as important sources of meaning in life (e.g., family, friends, relationships, [15, 34]). Thus, it would be important to examine whether interpersonal factors, such as one's expectations for interpersonal

relationships (i.e., interpersonal expectancies), might add to, and interact with, meaning in life to further predict variance in depressive symptoms differently for men and women. Taken together, one might hypothesize that the positive expectation of having close and supportive relationships (i.e., positive interpersonal expectancy) may strengthen the negative association between presence of meaning in life and depressive symptoms, and alternatively, may weaken any positive association that might exist between searching for meaning in life and depressive symptoms for women more so than for men. Moreover, the negative expectation of not being able to develop close and supportive relationships (i.e., negative interpersonal expectancy) may weaken the negative association between presence of meaning in life and depressive symptoms, while it may exacerbate any positive association that might exist between searching for meaning in life and depressive symptoms for women more so than for men. However, as noted above, gender differences in the association between meaning in life and depressive symptoms, much less any potential moderators (e.g., positive and negative interpersonal expectancies) that might impact that association for men and women, have not been studied and are left to be investigated further.

Purpose of the Present Study

We conducted the present study to (a) examine for gender differences in the correlations between meaning in life, interpersonal expectancies, and depressive symptoms; (b) determine whether meaning in life is a significant predictor of depressive symptoms in men and women; (c) examine whether interpersonal expectancies would add to the prediction of depressive symptoms for men and women; and (d) examine whether the dimensions of meaning in life would interact with the dimensions of interpersonal expectancies to further predict depressive symptoms in men and women beyond the main effects.

Similar to findings from past studies (e.g., [51]), we expected presence of meaning in life to be negatively associated with depressive symptoms and searching for meaning in life to be positively associated with depressive symptoms, with stronger associations found for women than for men. Additionally, given that interpersonal factors are important in relation to depressive symptoms, especially among women (e.g., [31]), we predicted interpersonal expectancies to be significantly related to depressive symptoms. Specifically, we expected to find positive interpersonal expectancy to be negatively associated with depressive symptoms and negative interpersonal expectancy to be positively associated with depressive symptoms. Furthermore, as meaning in life may be drawn from interpersonal sources (e.g., [34]), we predicted meaning in life and interpersonal expectancies to be significantly related to each other.

Consistent with past findings (e.g., [51, 55]) and with the notion that meaning in life represents an important determinant of adjustment, especially for women, we predicted meaning in life to account for a significant amount of variance in depressive symptoms for both groups, but for a greater amount of variance for women. Furthermore, we expected to find interpersonal expectancies to add to the

prediction of depressive symptoms, as interpersonal factors have been found to be important determinants of depressive symptoms [10]. Finally, as interpersonal factors and relationships are involved with both meaning in life and experiences of depression, we expected interpersonal expectancies to interact with meaning in life to further predict depressive symptoms above and beyond the main effects. However, given that women are often more motivated by interpersonal factors/relationships, we hypothesized that there may be a greater number of significant interactions involving meaning in life dimensions and interpersonal expectancies dimensions for women than for men.

Method

Participants

The study recruited a total of 258 (123 male and 135 female) students from a large Midwestern public university. Of the original 258 participants, 9 did not complete all study measures and thus the present study is based on a total of 249 (117 male and 132 female; self-reported) students. Participants' ages range from 18 to 29 with a mean age of 20.89 ($SD = 1.47$). Within the present sample, 7.4 % of participants are freshmen students, 9.7 % are sophomores, 32.9 % are juniors, 46.9 % are seniors, and 5 % are graduate students or other. Of the participants, 80.6 % reported to be Caucasian American, 10.5 % reported to be Asian American/Pacific Islander, 3.1 % reported to be Hispanic/Latino, 1.9 % reported to be African American, 1.9 % reported to be biracial/mixed, and 1.6 % reported to be other. Of the participants, 1.9 % considered their family to be lower class, 7.4 % lower-middle class, 25.2 % middle class, 48.8 % upper-middle class, and 16.3 % upper class. With regards to income, 84.7 % of participants reported household incomes of \$50,001 or more, 10.4 % of participants reported a household income between \$30,001 and \$50,000, 4 % of participants reported a household income between \$10,000 and \$30,000, and less than 1 % of participants reported a household income less than \$10,000.

Measures

Demographic Variables

The regression analyses included a number of demographic variables to act as controls in order to better compare findings between the male and female samples. Age, race, class, social class, and income were all self-reported by the participants. For race, the coding used is 1 for Caucasian American, 2 for African American, 3 for Hispanic/Latino, 4 for Asian American/Pacific Islander, 5 for Biracial/Mixed, and 6 for Other. For class, freshmen, sophomore, junior, senior, and other were coded 1–5, respectively. For social class, lower class, lower-middle class, middle class, upper-middle class, and upper class were coded 1–5, respectively. Finally,

each income bracket (see above), from less than \$10,000 to more than \$50,000 was coded 1–4, respectively.

Meaning in Life

Meaning in life was assessed for using the Meaning in Life Questionnaire (MLQ; [49]). The MLQ is composed of ten items, five of the total items represent presence of meaning in life and the remaining five items represent searching for meaning in life. An example item of Meaning in Life-Presence is “I have a good sense of what makes my life meaningful.” An example item of Meaning in Life-Searching is “I am looking for something that makes my life feel meaningful.” Each set of five items was summed to make up the two subscales of the MLQ. Participants are asked to rate how true each statement is to them using a 7-point Likert-type scale ranging from 1 (absolutely untrue) to 7 (absolutely true). In general, higher scores represent greater presence of meaning in life and greater searching for meaning in life. In the present study, internal reliability of Meaning in Life-Presence, as measured by Chronbach’s alpha, was .84 for males and .86 for females. The internal reliability for Meaning in Life-Searching was .80 for males and .90 for females.

Interpersonal Expectancies

Interpersonal expectancies was assessed using select items from the Children’s Optimistic and Pessimistic Expectations of Relationships Scale (COPER; [38]). While the COPER was developed for use with children, the original 16-item self-report measure includes eight items that are specific to expectations of future relationships with kids as well as eight items that are more general statements regarding general future expectations with people. The present study only included the eight items about relationship expectancies with people. Other existing measures of relationship expectancies are unipolar, including only positive or negative expectations for relationships, or they are solely focused on specific relationships (e.g., romantic partners) rather than more general relationships. Of the select eight items we included for this study, four items make up the positive interpersonal expectancies dimension and four items make up the negative interpersonal expectancies dimension. An example of a positive interpersonal expectancy item is “In the future, people will care about me.” An example of a negative interpersonal expectancy item is “In the future, people will treat me badly.” The four positively worded items were summed to create the positive interpersonal expectancies subscale, and the four negatively worded items were summed to create the negative interpersonal expectancies subscale. Participants are asked to rate how true each statement is to them using a 5-point Likert-type scale ranging from 1 (not at all true) to 5 (very true). In general, greater scores indicate greater positive/negative expectations for future relationships. In the present study, internal reliability of positive interpersonal expectancies, as measured by Chronbach’s alpha, was .86 for males and .89 for females. The internal reliability for Negative Interpersonal Expectancies was .79 for males and .87 for females.

Depressive Symptoms

Depressive symptoms was measured using the 13-item Beck Depression Inventory [6]. This scale is a 13-item self-report measure of depressive symptomatology. Participants are asked to choose the option that best describes their current condition using a 4-point Likert-type scale. For example “(0) I do not feel sad, (1) I feel sad, (2) I am sad all the time and I can’t snap out of it, (3) I am so sad and unhappy that I can’t stand it.” The thirteen items were summed to create a total depressive symptoms score. In general, higher scores on the 13-item BDI represent greater depressive symptoms. In the present study, internal reliability of the 13-item Beck Depression Inventory, as measured by Chronbach’s alpha, was .89 for males and .92 for females.

Procedure

Approval for the study was obtained from the Institutional Review Board prior to data collection. Participants were recruited from an upper-level psychology course. Participants were offered extra credit for the course for completion of the survey. Each participant received an unsealed and unmarked plain white envelope containing the survey. Participants were asked to complete the survey between class meetings and to seal their survey response with responses in the envelope included. Participant information remained strictly confidential as no identifiable information was collected.

Results

Means and standard deviations for all study measures and control variables are presented in Table 1 for males and females (note: no significant gender differences were found on the mean levels of any of the study measures). Correlations for all study measures and control variables are presented in Table 2 with bold values representing correlations of study variables for females. As expected, presence of meaning in life was significantly and negatively associated with depressive symptoms for both males and females ($r_s = -.34$ and $-.57$, $p < .001$, respectively). The difference between the two correlation coefficients was significant ($z = 2.28$, $p = .02$). Interestingly, searching for meaning in life was only found to be significantly correlated with depressive symptoms for females ($r = .31$, $p < .001$). However, given the sample size differences between the two groups, this difference was not found to be significant ($z = -1.32$, $p = .19$). Furthermore, positive interpersonal expectancy was found to be significantly and negatively associated with depressive symptoms for males and females ($r_s = -.45$ and $-.52$, $p < .001$, respectively), and negative interpersonal expectancy was found to be significantly and positively associated with depressive symptoms for both males and females ($r_s = .29$ and $.38$, $p < .001$, respectively). Notably, both positive and negative interpersonal expectancies were found to be significantly associated with presence of meaning in life for both males and females ($r_s = -.29$ to $.45$,

Table 1 Means and standard deviations of all study variables by gender

Study variable	Male (<i>n</i> = 117)		Female (<i>n</i> = 132)	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Age	21.05	1.73	20.81	1.17
Class	3.33	1.02	3.29	.83
Social class	3.56	.89	3.79	.87
Income	3.84	.41	3.78	.60
Meaning in life—presence	23.94	7.93	23.80	6.30
Meaning in life—searching	23.37	6.08	24.77	6.44
Positive interpersonal expectancy	17.29	2.64	17.48	2.73
Negative interpersonal expectancy	8.69	3.19	9.23	3.58
Depressive symptoms	4.46	5.64	5.25	6.29

$p < .001$), however, neither dimensions of interpersonal expectancy were found to be significantly associated with searching for meaning in life.

To examine the predictive utilities of meaning in life and interpersonal expectancies in accounting for variance in depressive symptoms, we conducted hierarchical regression analyses separately for males and females (Tables 3 and 4, respectively). All assumptions for conducting regression analyses were met. Demographic variables (i.e., age, race, class, social class, and income) may contribute to differences found in experiences of depressive symptoms as well as experiences of meaning in life and interpersonal expectancies. Thus, to account for these potential influences, demographic variables were entered as a set in the First Step as controls. Meaning in life (i.e., presence and searching) was entered as a set in the Second Step. As previously noted, interpersonal factors, particularly positive and negative expectations for interpersonal relationships, may buffer and/or exacerbate the relationship between meaning in life and depressive symptoms. Therefore, interpersonal expectancies (i.e., positive and negative) was included as the Third Step, and the multiplicative terms of meaning in life $\times$ interpersonal expectancies were included in the Fourth Step. To determine if meaning in life, interpersonal expectancies, and the interactions accounted for a small, medium, or large amount of the variance in depressive symptoms, we used Cohen's [11] convention for small ($f^2 = .02$), medium ($f^2 = .15$), and large effects ($f^2 = .35$).

As shown in Table 3, the demographic variables were not found to account for a significant amount of variance in depressive symptoms in males, $F(5, 111) = 1.52$, $p = .19$. When meaning in life was entered in Step 2, it was found to account for an additional medium ($f^2 = .19$), 16 % of variance in depressive symptoms, $F(2, 109) = 11.20$, $p < .001$. Specifically, of the two dimensions of meaning in life, only presence of meaning in life emerged as a significant individual predictor of depressive symptoms in males ($\beta = -.38$, $p < .001$). Furthermore, when interpersonal expectancies was added in the Third Step, it

Table 2 Correlations between all study measures in males ($n = 117$) and females ($n = 132$)

Measures	1	2	3	4	5	6	7	8	9
1. Age	–								
	–								
2. Race	–.11	–							
	.56	–							
3. Class	.68***	.23*	–						
	.74***	.07	–						
4. Social class	–.09	.03	–.06	–					
	–.09	–.07	–.09	–					
5. Income	–.02	.16	–.03	.48***	–				
	–.06	.04	–.07	.62***	–				
6. Presence	.11	.04	.10	–.10	–.15	–			
	.01	–.07	.18*	.04	.09	–			
7. Searching	–.07	–.11	–.02	–.13	–.12	–.02	–		
	–.08	.02	–.05	.05	.03	–.22*	–		
8. Positive interpersonal expectancy	.05	.15	.06	–.01	.16	.42***	–.08	–	
	.01	–.13	.04	.02	–.01	.38***	–.16	–	
9. Negative interpersonal expectancy	.04	.05	.10	–.04	–.23**	–.20*	.09	–.27**	–
	–.02	–.07	–.03	–.03	.07	–.29***	.13	–.44***	–
10. Depressive symptoms	.07	.07	–.03	–.07	–.20*	–.32***	.15	–.45***	.20**
	–.04	.07	–.08	–.22**	–.20*	–.57***	.31***	–.52***	.38***

Numbers in bold represent correlational values from the female sample

* $p < .05$; ** $p < .01$; *** $p < .001$

was found to account for a small ($f^2 = .11$), but significant, 10 % of additional variance in depressive symptoms, $F(2, 107) = 7.70$, $p < .001$. Specifically, only positive interpersonal expectancy emerged as a significant individual predictor of depressive symptoms for males ($\beta = -.34$, $p < .001$). Finally, when the interactions were added to Step 4, it was found to account for a small ($f^2 = .09$), but significant 8 % of additional variance in depressive symptoms for males $F(4, 103) = 3.47$, $p = .011$. Specifically, only the interaction term of Presence $\times$ Positive Interpersonal Expectancy emerged as a significant individual predictor of depressive symptoms for males ($\beta = 1.61$, $p = .013$).

As shown in Table 4, the demographic variables were also not found to account for a significant amount of variance in depressive symptoms in females, $F(5, 123) = 1.98$, $p = .087$. When meaning in life was entered in Step 2, it was found to account for an additional large ($f^2 = .49$), 33 % of variance in depressive symptoms, $F(2, 121) = 33.55$, $p < .001$. Specifically, both presence of meaning

Table 3 Results of hierarchical regression analyses showing amount of variance in depressive symptoms accounted for by meaning in life and interpersonal expectancies in males (N = 117)

Outcome and measure	β	R^2	ΔR^2	F
Depressive symptoms				
Step 1: Demographics		.06	–	1.52
Age	.16			
Race	.10			
Class	–.11			
Social class	.05			
Income	–.24*			
Step 2: Meaning in life		.22	.16	11.20***
Presence	–.38***			
Searching	.14			
Step 3: Interpersonal expectancies		.32	.10	7.70***
Positive interpersonal expectancy	–.34***			
Negative interpersonal expectancy	.09			
Step 4: Interactions		.40	.08	3.47*
Presence $\times$ Positive Interpersonal Expectancy	1.61*			
Presence $\times$ Negative Interpersonal Expectancy	–.68			
Searching $\times$ Positive Interpersonal Expectancy	.46			
Searching $\times$ Negative Interpersonal Expectancy	.12			

Standardized β s are reported* $p < .05$; *** $p < .001$

in life ($\beta = -.52$, $p < .001$) and searching for meaning in life ($\beta = .21$, $p = .004$) emerged as significant individual predictors of depressive symptoms in females. The regression coefficients for presence of meaning in life for males and females are significantly different $t(245) = 2.68$, $p = .008$, however the regression coefficients for searching for meaning in life for males and females are not significantly different $t(245) = .72$, $p = .469$. Furthermore, when interpersonal expectancies was added in the Third Step, it was found to account for a small ($f^2 = .12$), but significant 11 % of additional variance in depressive symptoms, $F(2, 119) = 13.12$, $p < .001$. Specifically, only positive interpersonal expectancy emerged as a significant individual predictor of depressive symptoms for females ($\beta = -.29$, $p < .001$). Finally, when the interaction terms were added to Step 4, they was found to account for a small ($f^2 = .09$), but significant 8 % of additional variance in depressive symptoms for females, $F(4, 115) = 5.97$, $p < .001$. Of the four possible interaction terms, three interactions emerged as significant individual predictors of depressive symptoms in females, specifically Presence $\times$ Positive Interpersonal Expectancy ($\beta = 2.44$, $p < .001$), Presence $\times$ Negative Interpersonal Expectancy ($\beta = .96$, $p = .008$), and Searching $\times$ Positive Interpersonal Expectancy ($\beta = -1.11$, $p = .047$).

Table 4 Results of hierarchical regression analyses showing amount of variance in depressive symptoms accounted for by meaning in life and interpersonal expectancies in females (N = 132)

Outcome and measure	β	R^2	ΔR^2	F
Depressive symptoms				
Step 1: Demographics		.07	–	1.98
Age	.05			
Race	.07			
Class	–.15			
Social class	–.15			
Income	–.12			
Step 2: Meaning in life		.41	.33	33.55***
Presence	–.52***			
Searching	.21**			
step 3: interpersonal expectancies		.51	.11	13.12***
Positive interpersonal expectancy	–.29***			
Negative interpersonal expectancy	.13			
Step 4: Interactions		.60	.08	5.97***
Presence $\times$ positive interpersonal expectancy	2.44***			
Presence $\times$ negative interpersonal expectancy	.96**			
Searching $\times$ positive interpersonal expectancy	–1.11*			
Searching $\times$ negative interpersonal expectancy	.29			

Standardized β s are reported* $p < .05$; ** $p < .01$; *** $p < .001$

To visually inspect the manner in which meaning in life interacted with interpersonal expectancies in predicting depressive symptoms for men and women, we made three interaction plots based on our initial regression results. In Fig. 1, we plotted the regression of depressive symptoms on presence of meaning in life (shown at low and high levels, using a mean split of 23.45) for males at low and high levels of positive interpersonal expectancy (using a mean split of 17.29). In Fig. 2 Panel A, we plotted the regression of depressive symptoms on presence of meaning in life (mean split of 23.80) for females at low and high levels of positive interpersonal expectancy (mean split of 17.48). Panel B shows the same regression for females at low and high levels of negative interpersonal expectancy (mean split of 9.23). Finally, in Fig. 3, we plotted the regression of depressive symptoms on searching for meaning in life (mean split 24.77) for females at low and high levels of positive interpersonal expectancies.

As shown in Fig. 1, level of depressive symptoms for men is higher among those with low presence of meaning in life than for those with high presence of meaning in life. However, level of depressive symptoms appears to be much lower for men with high positive interpersonal expectancy compared to low positive interpersonal expectancy. This relationship is similar and even more apparent among women. Figure 2a shows that levels of depressive symptoms are lower for women with high compared to low positive interpersonal expectancy at both low and high levels of

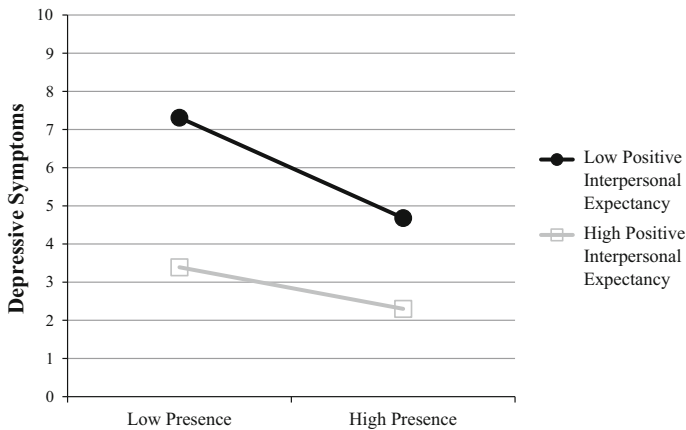


Fig. 1 Depressive symptoms at low versus high levels of positive interpersonal expectancy for males with low presence and high presence of meaning in life. *Note* Scale does not illustrate the full possible range of scores on the BDI-13 (i.e., 0–39)

meaning in life. This indicates that positive interpersonal expectancies strengthen the inverse relationship between presence of meaning in life and depressive symptoms. Furthermore, Fig. 2b shows that women with high negative interpersonal expectancy have much higher levels of depressive symptoms than those with low negative interpersonal expectancy at both low and high levels of meaning in life. This suggests that negative interpersonal expectancies weaken the inverse relationship between presence of meaning in life and depressive symptoms. Finally, in Fig. 3, the positive relationship between searching for meaning in life and depressive symptoms is not as strong among women with high positive interpersonal expectancy, compared to those with low positive interpersonal expectancy. Thus, this supports the notion that having high levels of positive interpersonal expectancy weakens the positive relationship that exists between searching for meaning in life and depressive symptoms among women.

Discussion

Meaning in life has been argued to be an important construct involved with adjustment outcomes such as depressive symptoms [51], however, it is not clear whether there may be potential gender differences in this relationship. Additionally, it has been suggested that men and women differ in how interpersonal factors affect mental health, including experiences of depression (e.g., [2, 31]). Thus, the present study sought to better understand how meaning in life and interpersonal expectancies may additively and interactively predict depressive symptoms differently for men and women.

Consistent with past findings [51], our correlational results indicated that presence of meaning in life was significantly and negatively associated with

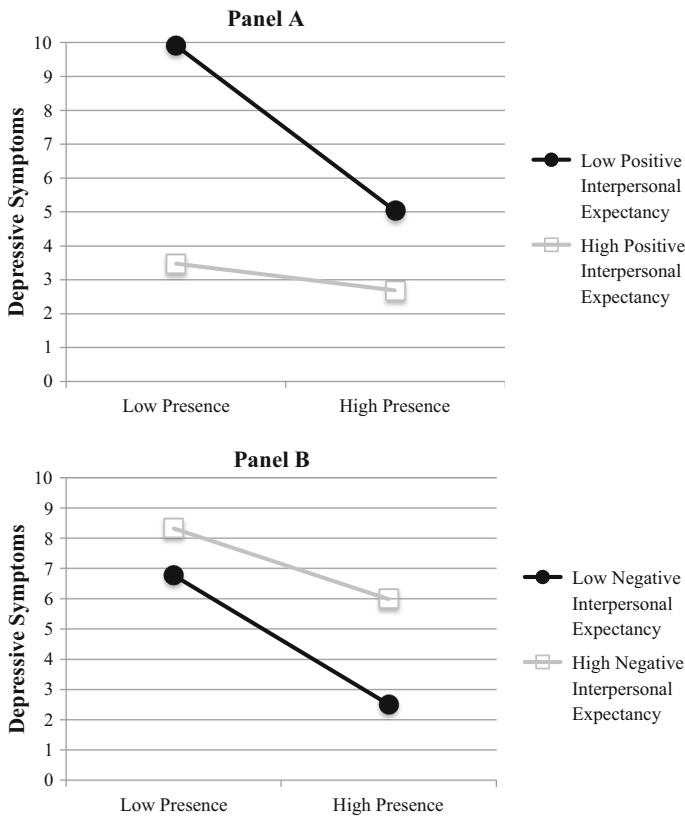


Fig. 2 *Panel A* Depressive symptoms at low versus high levels of positive interpersonal expectancy for females with low presence and high presence of meaning in life. *Panel B* Depressive symptoms at low versus high levels of negative interpersonal expectancy for females with low presence and high presence of meaning in life. *Note* Scale does not illustrate the full possible range of scores on the BDI-13 (i.e., 0–39)

depressive symptoms. As predicted, the association was stronger for women than for men, providing evidence for some gender differences in the relationship between meaning in life and depressive symptoms. Coupled with our lack of finding significant mean differences in these variables, our findings lend support for the need to extend research beyond simply exploring mean differences to further examine for group differences in the function of these variables. Moreover, we found evidence that interpersonal expectancies are significantly related to depressive symptoms for both men and women. Finally, we found interpersonal expectancies to be related to presence of meaning in life, but not searching for meaning in life. Future research may seek to further examine this difference between presence of meaning in life and searching for meaning in life. For example, it may be that interpersonal factors/relationships serve as sources for one's perception of meaning in life, but they do not serve as important motivators for searching for meaning in life.

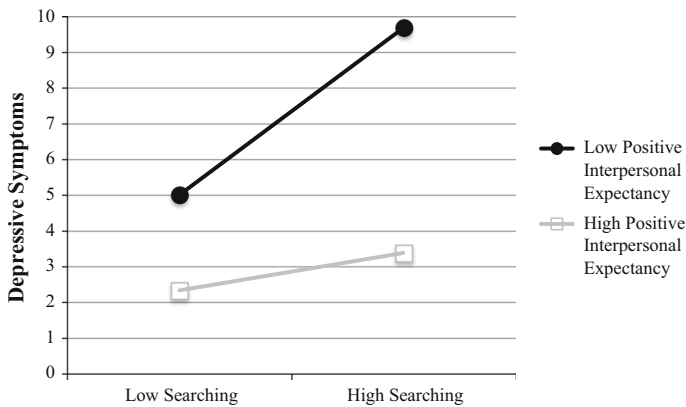


Fig. 3 Depressive symptoms at low versus high levels of positive interpersonal expectancy for females with low searching and high searching for meaning in life. *Note* Scale does not illustrate the full possible range of scores on the BDI-13 (i.e., 0–39)

Furthermore, this study sought to examine whether gender differences may exist in how meaning in life and interpersonal expectancies function as additive and interactive predictors of depressive symptoms. The findings support that meaning in life accounted for a greater amount of variance in depressive symptoms for women than for men. As the differences between males and females in the regression coefficients was significant for presence of meaning in life but not for searching for meaning in life, the findings indicate that the function of searching for meaning in life in relation to depressive symptoms may be consistent for both genders, however the role of presence of meaning in life in relation to depressive symptoms is more robust for women compared to men. One explanation for these findings is that given the shared experiences of exploration through emerging adulthood in college, the function of searching for meaning in life in the present sample is equally relevant in relation to experiences of depressive symptoms for men and women [3]. However, given some differences in the goals of college women compared to college men (e.g., being well educated, having close friends, and having a lot of relatives/family; [1]), presence of meaning in life may have a more robust role in relation to depressive symptoms for women. For example, it may be that greater aspirations for, or attainment of, certain goals, such as having close friends, provide women with a greater sense of meaning in life that works to better protect against depressive symptoms compared to men. On the other hand, if women perceive particular interpersonal goals as unattainable, it may be more difficult for them to perceive meaning in life and may lead to greater experiences of depressive symptoms than if men experienced similar challenges pertaining to interpersonal goals. While the results of this study are preliminary, one possible implication of these results may be that meaning based interventions that seek to increase perceptions of meaning in life may be especially useful for mitigating the experiences of depressive symptoms for women.

To further build off the findings of this study, it may be important for future studies to examine the different sources from which men and women derive

meaning in life [47], and to examine whether there may be gender differences in how different sources of meaning in life predict different adjustment outcomes. Indeed, researchers have found that men and women differ in sources of other positive constructs. For example, Reid [43] found that for men, self-esteem is an important source of positive well-being, but for women, sources of positive well-being include self-esteem and relationship harmony. Moreover, it may also be important to examine whether there are particularly harmful methods used or certain contexts in which searching for meaning in life may exacerbate depressive symptoms. For example, in a longitudinal study of female breast cancer patients, Kernan and Lepore [32] found that women who engage in greater searching for meaning were more likely to experience greater negative affect over time. Thus, continuous and unresolved searching for meaning following negative events may involve greater distress and may be increasingly frustrating due to the inability to find what is sought after. Notably, however, searching for meaning is not simply the absence of meaning in one's life. Indeed, Steger et al. [50] argued that there may be many different motivations for searching for meaning in life (e.g., deficits in well-being, tendency to seek out novelty, pursuit of future goals, inability to let go of painful past experiences, etc.), and urge for more research in order to better understand the complexities of how searching for meaning is related to well-being and personal functioning. Taken together, by identifying the sources of meaning in life as well as the challenges in, and motivations for, the search to attain meaning in life, researchers will be able to better identify predictors of depressive symptoms that may be specific to men or women and predictors that are common to both groups.

Additionally, support was found for interpersonal expectancies as an important statistical predictor of depressive symptoms above and beyond meaning in life. Common to both men and women, the present findings show that positive interpersonal expectancy was inversely related with depressive symptoms. Thus, there may be utility in interventions to mitigate depressive symptoms for men and women by focusing on increasing the belief that one's future relationships will be good and supportive. Notably, in a study of perceived social support of undergraduate students, Tinajero et al. [53] found that female students perceived greater social support, and had larger social networks compared to males. Indeed, theories of gender differences in social support seeking [4, 14, 19] have argued that values of interpersonal relationships and interpersonal skills are more salient among women compared to men which may contribute to greater social support seeking by women when dealing with life problems or stressors. Thus, interventions to increase positive expectations of future relationships may be more effective for women, who may be able to draw evidence from existing larger social networks in order to change unrealistic expectations of future relationships.

Combining efforts to increase positive expectancies of future relationships with efforts to increase the perception of meaning in life may be especially useful for women. In the present study, a greater number of significant interactions were found for women compared to men indicating that the model for predicting depressive symptoms with these constructs appears to be much more complex for women than for men. As previously noted, there may be some factors specifically involved with

being a woman that make the perception of, and search for, meaning in life particularly challenging. Specifically, while preliminary, our findings may suggest that having greater negative expectations, or fewer positive expectations, for future relationships may make perceiving meaning and the process of searching for meaning in life especially difficult for women which may increase the likelihood of experiencing depressive symptoms. Accordingly, when assessing for depressive symptoms in women, it may also be important to assess for women's sense of meaning in life and interpersonal expectancies and note that women who have low levels of meaning in life and high negative expectations (or low levels of positive expectations) for future relationships may be at particularly high risk for depression. The findings of the present study provide a foundation for future research to further examine whether combining intervention efforts that target both meaning in life and interpersonal expectancies may combat depressive symptoms for women. Working together, increases in levels of meaning in life may help women to have greater positive expectations for their future relationships, and vice versa, having greater positive expectations for future relationships may also increase a women's perception of meaning in her life. The findings from the present study point to the need to better understand gender-related factors that are associated with one's sense of meaning, expectations for future relationships, and experiences of depressive symptoms.

Some Limitations of the Present Study

Although the present findings provide promising preliminary empirical support for the integrative model involving meaning in life and interpersonal expectancies as additive and interactive factors involved in depressive symptoms, especially for women, some important limitations should be noted. First, given the cross-sectional nature of the present study, cause and effect cannot be determined. Indeed, it is possible that the relationships between meaning in life, interpersonal expectancies, and depression are bidirectional or reversed. In that regard, it would be important for future longitudinal studies to examine the directionality of these relationships as well as how meaning in life and interpersonal expectancies may predict changes in depressive symptoms over time. Second, the present study utilized a college sample of young adults. Thus, it would be important to replicate this study among other age groups to test the generalizability of these findings. Third, and relatedly, the racial/ethnic composition of the present sample is predominantly European American. Therefore, it is unclear whether these findings may be generalizable to specific racial/ethnic groups. Future studies may want to test this model in specific groups in order to examine for potential racial/ethnic group differences in how meaning in life and interpersonal expectancies are involved with depressive symptoms. Fourth, the measurement of gender used in the present study had limitations in that the survey asked participants to indicate "sex" rather than "gender." This is problematic given that sex is indicative of biological differences while gender encompasses cultural and social meanings. Relatedly, the survey only included two options (i.e., male and female) for participants to choose between. Thus, future studies should include more

gender options for participants to choose from (e.g., male, female, transgender male, transgender female, other, etc.) as well as measurements of gender roles, expectations, socialization, adherence to gender norms, etc. For example, Matud et al. [37] found that life satisfaction for women was highest for those who had greater adherence to female gender roles and social support, and for men, life satisfaction was highest for those who had greater adherence to male gender roles and self-esteem. Therefore, it may be important for future studies to also examine how socialization to gender role norms may influence men and women's sense of and search for meaning in life. Particularly, according to Matud et al. [37], forming social interactions is central to femininity; thus femininity may only be a source of life satisfaction for women who have good social support relationships. This also points to the need to further examine how femininity may be related to women's expectations for future relationships (i.e., whether or not they will be supportive) and how that may influence meaning in life. Finally, the present study only sought to examine for gender differences in meaning in life as a predictor of depressive symptoms. Given that Steger et al.'s [51] study included a wider range of both positive and negative outcomes, it would be important for future research to examine whether gender differences in the predictive utility of meaning in life also exist for other adjustment outcomes, both positive and negative.

Concluding Thoughts

In summary, we sought to extend Steger et al.'s [51] findings of significant gender differences in meaning in life by examining for differences in how meaning in life predicts depressive symptoms as well as to examine for interpersonal expectancies as a significant additive and interactive predictor of depressive symptoms above and beyond meaning in life for men and women. We found that meaning in life, particularly presence of meaning in life, was a significant predictor of depressive symptoms for both men and women, but a stronger predictor of depressive symptoms for women than for men. Additionally, we found that interpersonal expectancies added to the prediction of depressive symptoms for both men and women, and interacted with meaning in life to predict additional variance for both groups, especially for women. In conclusion, our study shows that meaning in life and expectancies for future relationships differentiates women's experiences of depressive symptoms from men's experiences.

Compliance with Ethical Standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. This article does not contain any studies with animals performed by any of the authors.

Informed Consent Informed consent was obtained from all individual participants included in the study.

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